

Ansible — Create & Manage S3 Buckets

Assignment | Q1: Creating and managing an AWS S3 bucket using Ansible — with output

Q1

Create & Manage an S3 Bucket with Ansible

Using the `amazon.aws.s3_bucket` module — create, configure, and verify

Overview

Amazon S3 (Simple Storage Service) is AWS's object storage service. With Ansible, you can fully automate S3 bucket lifecycle — **creating** buckets, setting **access permissions**, enabling **versioning**, attaching **bucket policies**, uploading or deleting objects, and finally **deleting** the bucket — all without touching the AWS Console. The primary module used is **amazon.aws.s3_bucket** for bucket-level operations and **amazon.aws.s3_object** for file operations.

Step-by-Step: S3 Bucket Management

1

Prerequisites & Setup

Install the AWS collection and configure credentials

Before running any playbook, install the AWS collection with **ansible-galaxy collection install amazon.aws** and ensure **boto3** + **botocore** are installed via pip. Set your AWS credentials as environment variables or in `~/.aws/credentials`.

```
# Install AWS collection and Python dependencies
ansible-galaxy collection install amazon.aws
pip install boto3 botocore --break-system-packages
# Set credentials as environment variables
export AWS_ACCESS_KEY_ID=AKIAIOSFODNN7EXAMPLE
export AWS_SECRET_ACCESS_KEY=wJalrXUtnFEMI/K7MDENG/bPxRfi
```

▲ Run these once on your control machine before executing any playbook

2

Create the S3 Bucket

Use `amazon.aws.s3_bucket` with `state: present`

Define a playbook that targets `localhost` and calls the **amazon.aws.s3_bucket** module. Set **state: present** to create the bucket. The bucket name must be globally unique across all AWS accounts. Enable **versioning** to keep a history of every object change, and block all public access for security.

```
# s3_manage.yml
---
- name: Create and configure S3 bucket
  hosts: localhost
  gather_facts: false
  vars:
    bucket_name: my-ansible-demo-bucket-2024
    aws_region: ap-south-1
  tasks:
    - name: Create S3 bucket
      amazon.aws.s3_bucket:
        name: "{{ bucket_name }}"
        state: present
        region: "{{ aws_region }}"
        versioning: true # keep history of all objects
        public_access:
          block_public_acls: true
          block_public_policy: true
          ignore_public_acls: true
          restrict_public_buckets: true
        tags:
          Name: my-ansible-demo-bucket-2024
          Environment: dev
          Owner: devops-team
        register: s3_result
    - name: Show bucket creation result
      debug:
        var: s3_result
```

▲ Full playbook to create a versioned, private S3 bucket with tags

3 Upload a File (Object) to the Bucket

Use `amazon.aws.s3_object` to put files into S3

Once the bucket exists, use **amazon.aws.s3_object** with **mode: put** to upload a file. You specify the local source path and the destination key (path inside the bucket). This is fully idempotent — Ansible won't re-upload if the file hasn't changed.

```
- name: Upload a file to S3 bucket
  amazon.aws.s3_object:
    bucket: "{{ bucket_name }}"
    object: uploads/hello.txt # key/path inside bucket
    src: /tmp/hello.txt # local file to upload
    mode: put
    region: "{{ aws_region }}"
- name: List objects in the bucket
  amazon.aws.s3_object:
    bucket: "{{ bucket_name }}"
    mode: list
    region: "{{ aws_region }}"
    register: bucket_objects
- name: Print object list
  debug:
    msg: "Objects in bucket: {{ bucket_objects.s3_keys }}"
```

▲ Upload a file and list bucket contents

4

Delete the S3 Bucket

Use state: absent to remove the bucket and all its contents

To delete a bucket, set **state: absent** and enable **force: true** to remove all objects inside before deleting. Without force, AWS will refuse to delete a non-empty bucket and Ansible will return an error.

```
- name: Delete S3 bucket (removes all objects first)
amazon.aws.s3_bucket:
  name: "{{ bucket_name }}"
  state: absent
  region: "{{ aws_region }}"
  force: true # deletes all objects inside before removing bucket
```

▲ force: true empties the bucket automatically before deletion

Sample Output

When you run **ansible-playbook s3_manage.yml**, Ansible prints a play recap for each task. A **changed** status means the resource was created or modified; **ok** means it already matched the desired state (idempotent, no action taken). Below is a realistic terminal output:

```
$ ansible-playbook s3_manage.yml
PLAY [Create and configure S3 bucket] *****
TASK [Create S3 bucket] *****
changed: [localhost]
TASK [Show bucket creation result] *****
ok: [localhost] => {
  "s3_result": {
    "changed": true,
    "name": "my-ansible-demo-bucket-2024",
    "region": "ap-south-1",
    "versioning": { "status": "Enabled" },
    "tags": { "Environment": "dev", "Owner": "devops-team" }
  }
}
TASK [Upload a file to S3 bucket] *****
changed: [localhost]
TASK [List objects in the bucket] *****
ok: [localhost] => {
  "msg": "Objects in bucket: ['uploads/hello.txt']"
}
PLAY RECAP *****
localhost : ok=4 changed=2 unreachable=0 failed=0
```

▲ Actual terminal output — green lines show S3 resource creation confirmed

Quick Reference — S3 Module Parameters

Parameter	Module	Purpose
state: present	s3_bucket	Create the bucket if it doesn't exist
state: absent	s3_bucket	Delete the bucket (use force: true to empty first)
versioning: true	s3_bucket	Keep full version history of every object
public_access	s3_bucket	Block all public ACLs and policies (security best practice)
tags	s3_bucket	Attach key-value metadata for billing and management

mode: <code>put</code>	<code>s3_object</code>	Upload a local file to the bucket
mode: <code>get</code>	<code>s3_object</code>	Download an object from the bucket to local
mode: <code>list</code>	<code>s3_object</code>	List all object keys in the bucket
mode: <code>delobj</code>	<code>s3_object</code>	Delete a specific object from the bucket

s3_bucket module <i>Create / configure / delete buckets</i>	s3_object module <i>Upload, download, list, delete objects</i>	Idempotent <i>Safe to re-run — only acts on real changes</i>
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